

Session:18

1. Extract valid Indian phone numbers (re.findall)

Valid: 10 digits starting with 7, 8, or 9

```
import re
text = "Call me at 9876543210 or 8123456789. Ignore 1234567890 and 9987654321."

pattern = r'\b[789]\d{9}\b'
phones = re.findall(pattern, text)

print(phones)
```

2. Check valid date using re.search (DD/MM/YYYY)

```
import re

def contains_date(text):
    pattern = r'\b\d{2}/\d{2}/\d{4}\b'
    return bool(re.search(pattern, text))

# Example
print(contains_date("Meeting on 25/06/2024")) # True
print(contains_date("No date here"))          # False
```

3. Extract email addresses from messy text (re.findall)

```
import re

text = """
Great food! contact us at info@zomato.com or support.food123@gmail.com.
Also reach billing@company.co.in for queries.
"""

pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
emails = re.findall(pattern, text)

print(emails)
```

4. Mask all but last 4 digits of phone numbers (re.sub only, no loops)

```
import re
```

```
text = "Paytm numbers: 9876543210 and 8123456789"

masked = re.sub(r'([789]\d{5})(\d{4})', r'*****\2', text)

print(masked)
```

Explanation:

- `([789]\d{5})` → first 6 digits
 - `(\d{4})` → last 4 digits (kept visible)
 - Replace first part with `*****`
-

5. Match Flipkart-style Order IDs

Typical format: `OD` + long number (example: `OD123456789012345000`)

Regex pattern:

```
pattern = r'\bOD\d{16,18}\b'
```

Test it:

```
import re

orders = [
    "Your order is OD123456789012345000 confirmed",
    "Invalid OD12345",
    "Track OD987654321098765432 status"
]

pattern = r'\bOD\d{16,18}\b'

for o in orders:
    match = re.search(pattern, o)
    print(o, "->", bool(match))
```